

Inpainting



Résultat attendus



Sommaire



1. Introduction
 - a. Etat de l'art
 - b. Jeu de données
2. Implémentation
 - a. Première approche MAE
 - b. Implémentation U-Net
 - c. Amélioration U-Net + GAN
 - d. Amélioration U-Net vers PEN-Net
3. Résultats
 - a. Unet + PatchGAN
 - b. FID Métriques
4. Interface (Adobe 3000)



Introduction

Etat de l'art

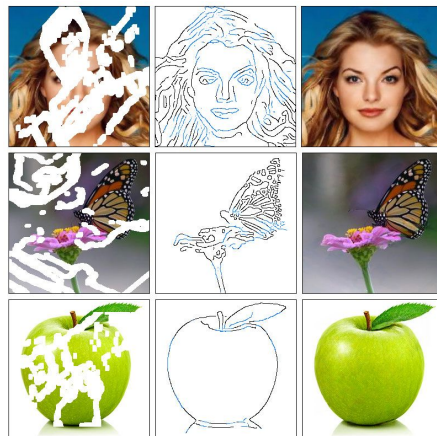
EdgeConnect (Nazeri et al. 2019)

Contextual Attention (Yu et al. 2018)

Resolution-robust Large Mask Inpainting with Fourier Convolutions (2021)

Aligned Stable Inpainting with UnKnown Areas Prior (ASUKA) (2023)

Learning Pyramid-Context Encoder Network for High-Quality Image Inpainting (PEN-Net) (2019)



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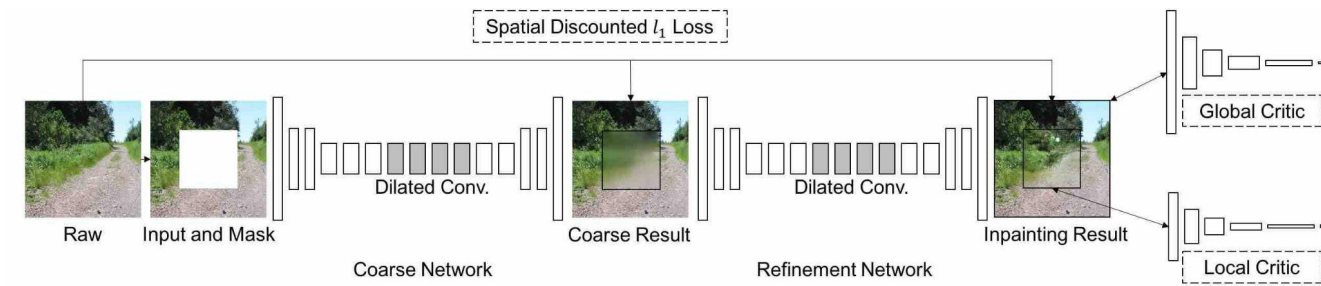
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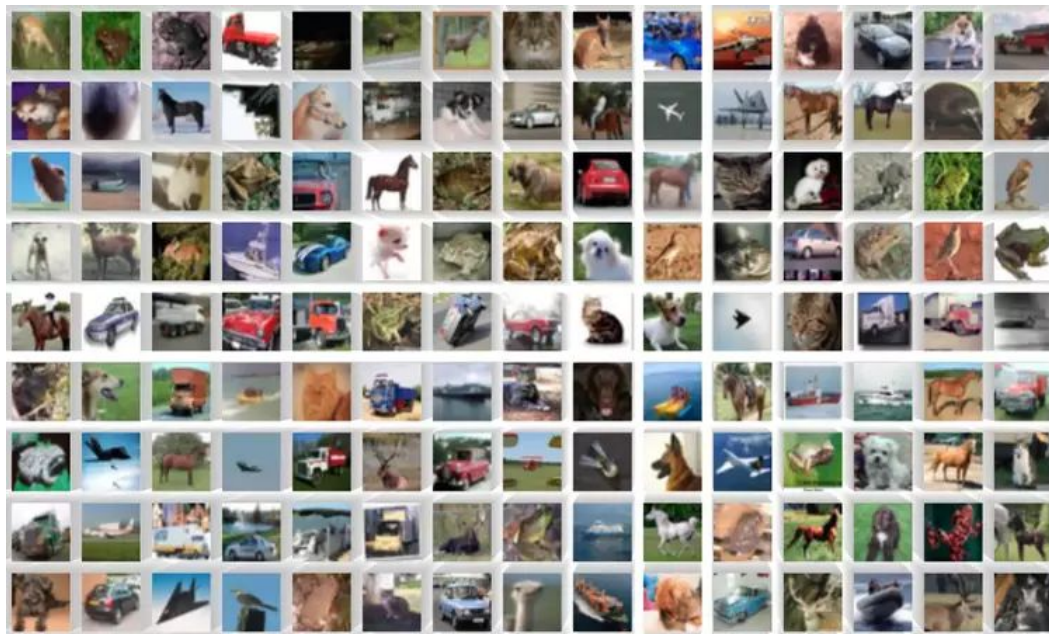
Learning Pyramid-Context Encoder Network for High-Quality Image Inpainting (PEN-Net) (2019)



Jeux de données

CIFAR 10 (32x32)
entraînement et test rapide

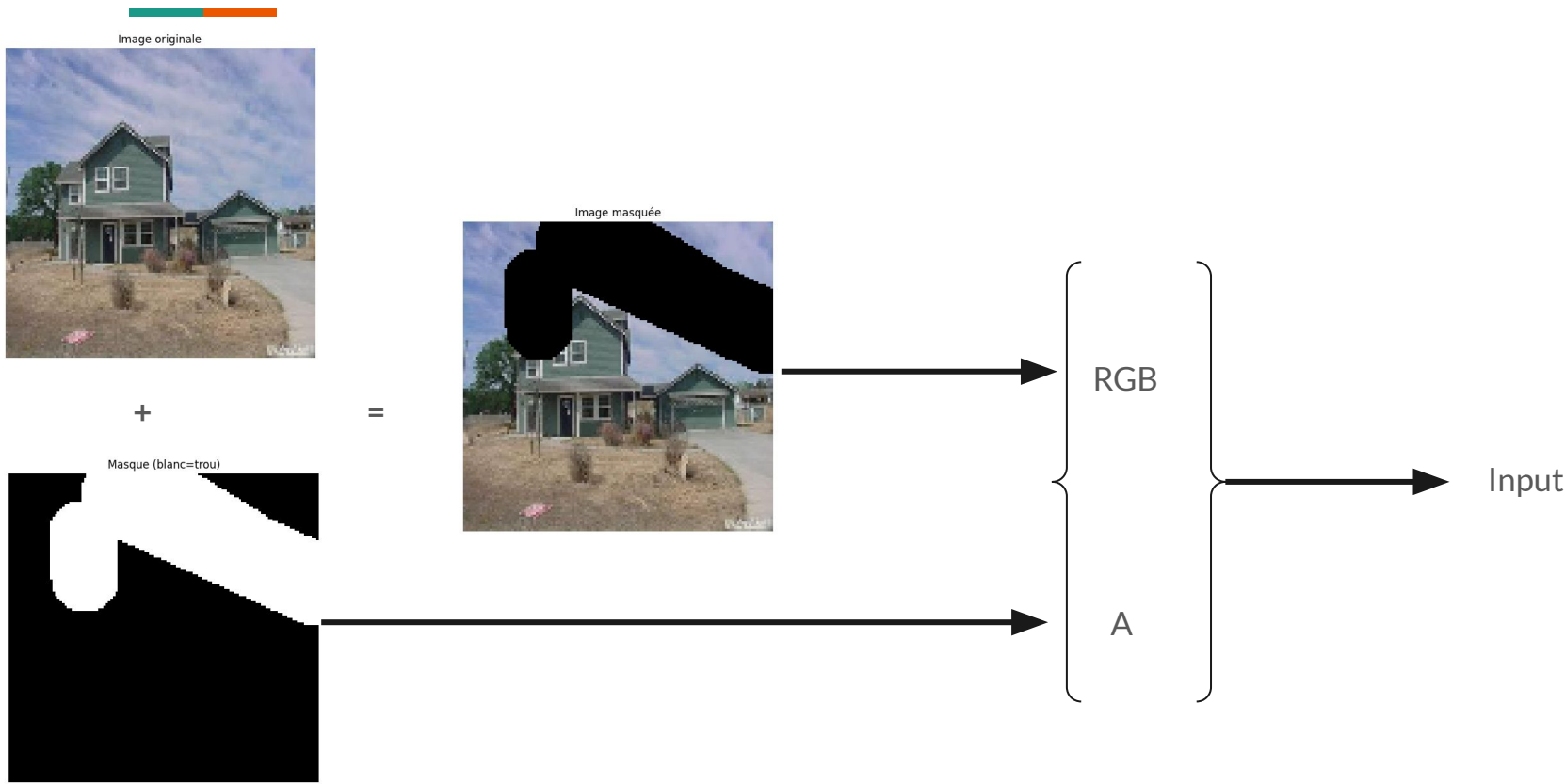
Places-2_MIT_Dataset (256x256)
plus détaillé



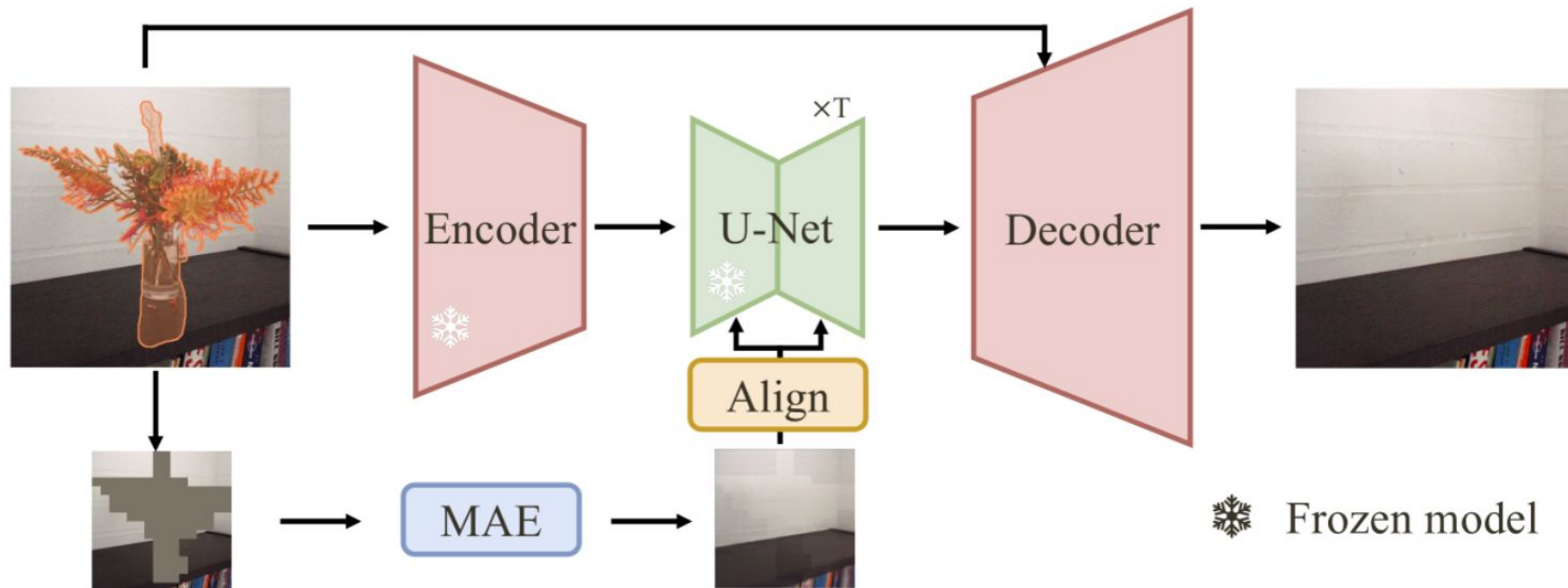


Implémentation

Entraînement

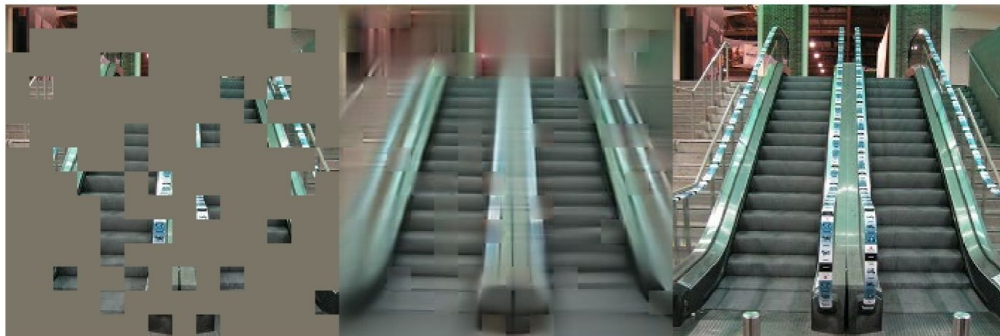


Aligned Stable Inpainting with UnKnown Areas Prior (ASUKA)



Difficultés rencontré MAE

Attendu:



Obtenu:



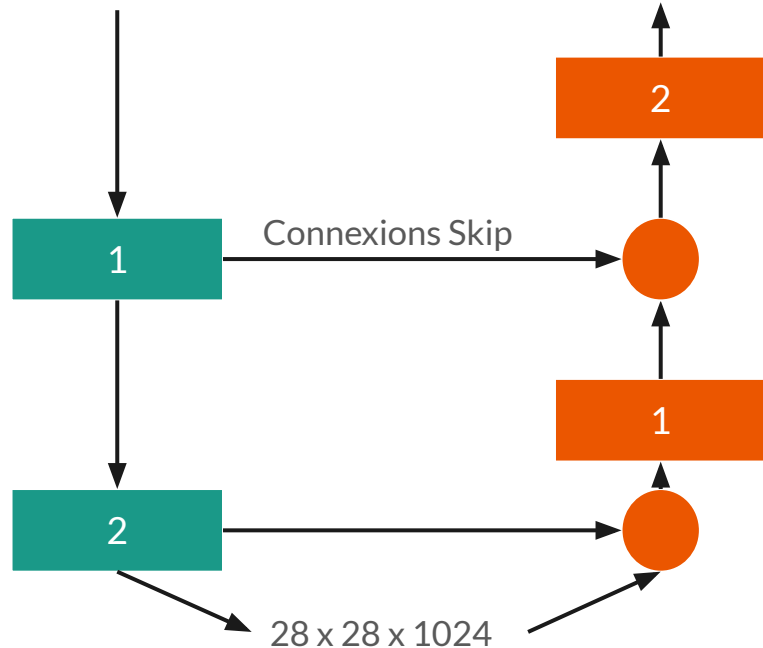


Nouveau départ : PEN-Net

U-Net



572 x 572 x 4



== Conv -> Conv (down) -> maxPool



== Conv -> Conv (up)



== Concaténation

Identification des features / patterns
transmise via "connexions skip"

Résultat obtenu avec U-Net

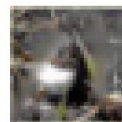
Image originale



Image masquée



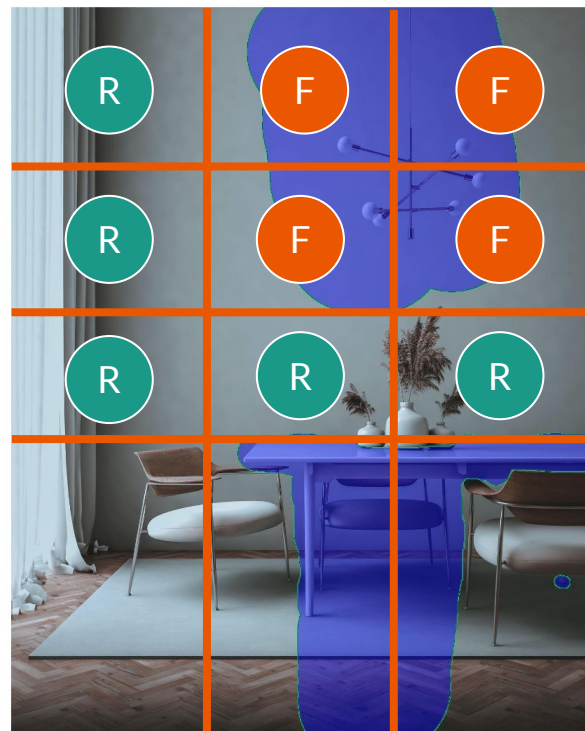
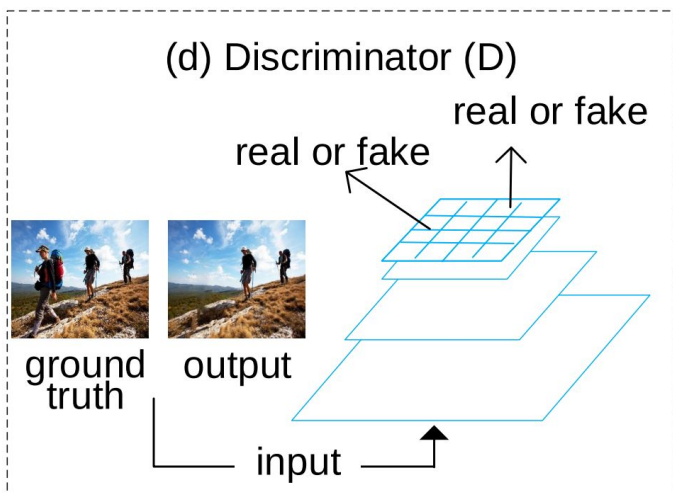
Image générée



Entraînement supervisé simple
Fonction de perte: MSE
250 epochs

Patch GAN - Discriminateur

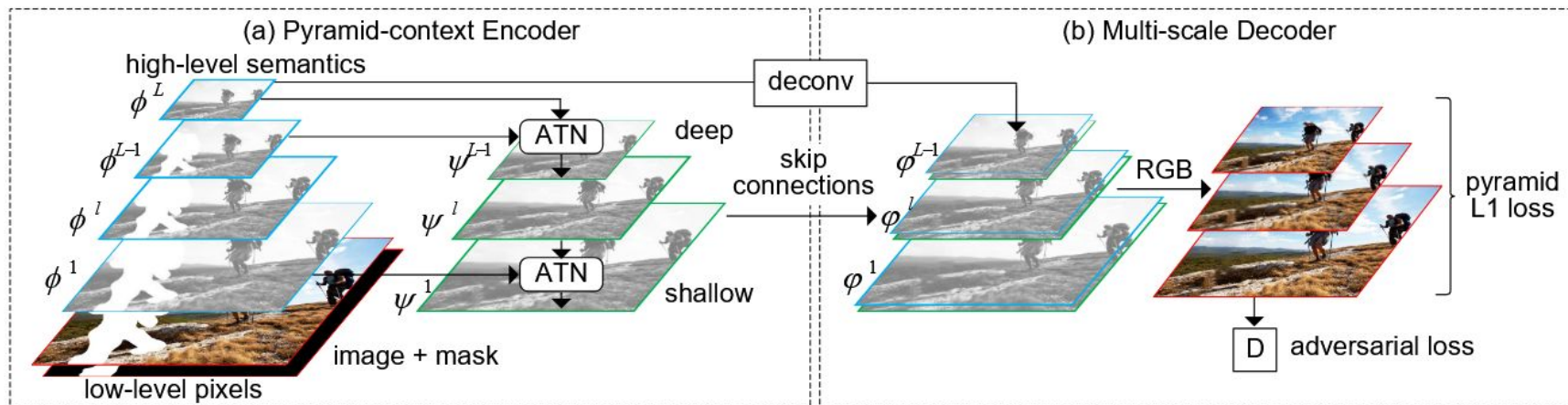
Accentue l'évaluation sur les zones masquées, restreint l'entraînement



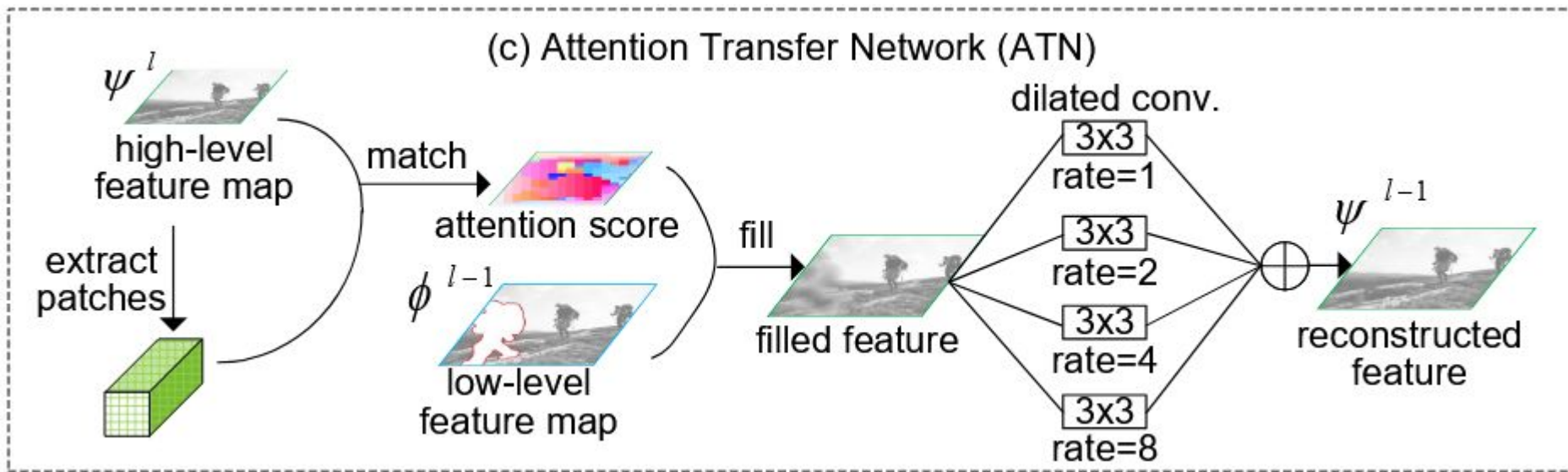


Unet vers PEN-Net

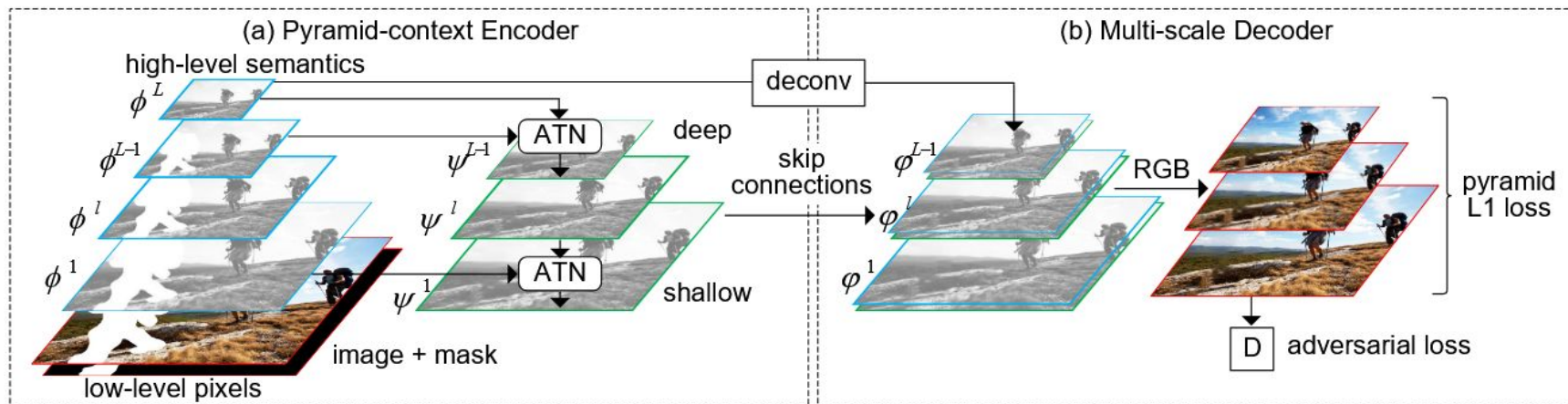
Pen-NET



Ajout d'Attention Transfer Network (ATN)



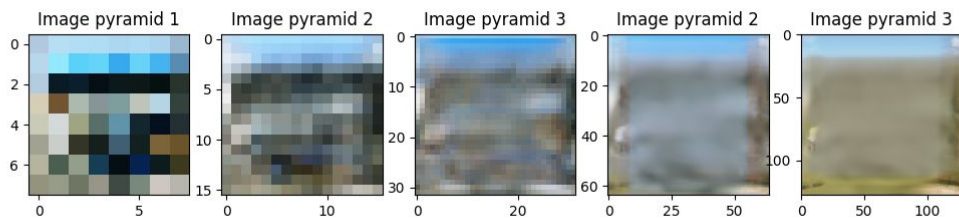
Pen-NET





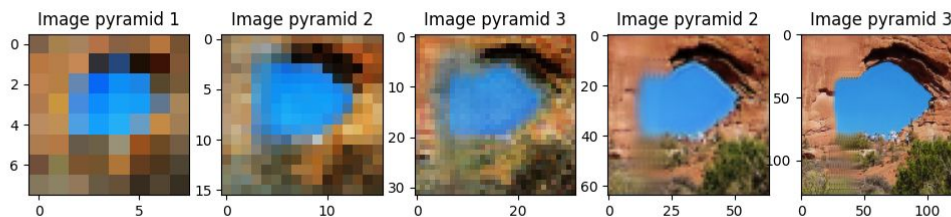
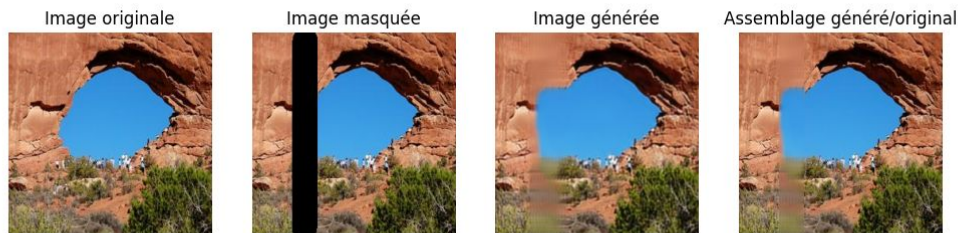
Résultats

Pen-NET (Premier test)



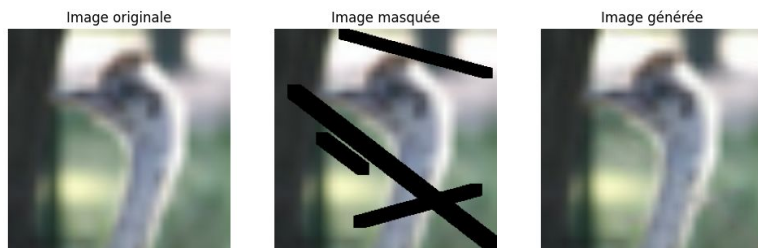
PSNR : 9.00 dB

Pen-NET (Après affinement des poids)



PSNR : 15.27 dB

U-Net GAN



CIFAR10
PSNR de 27.74 dB

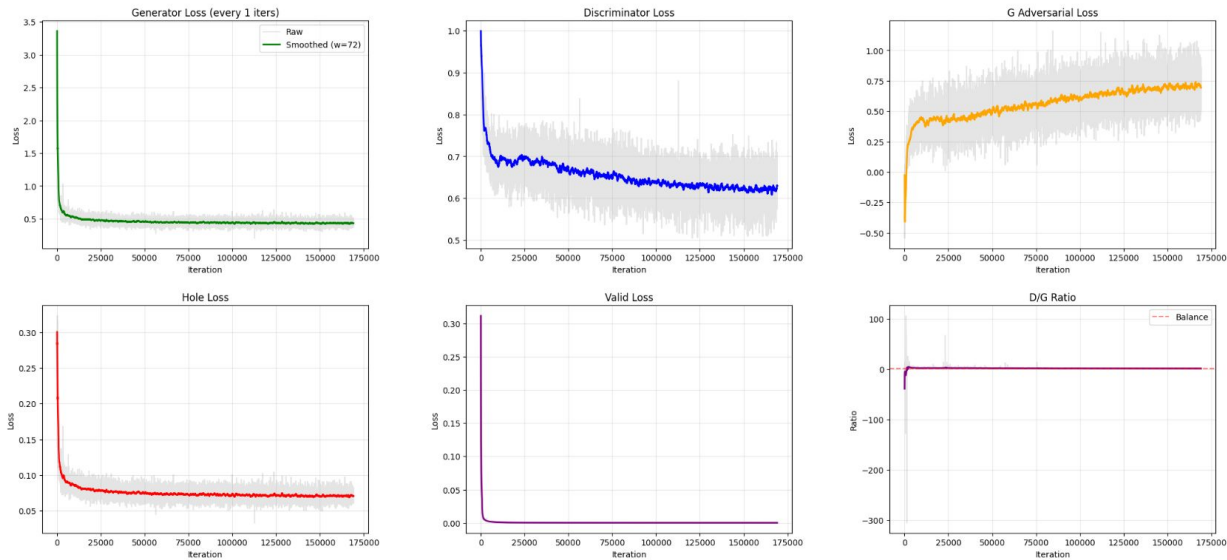


10% de Place2

U-Net GAN



Training Metrics - Showing 16909/16909 points



Plot every N iters

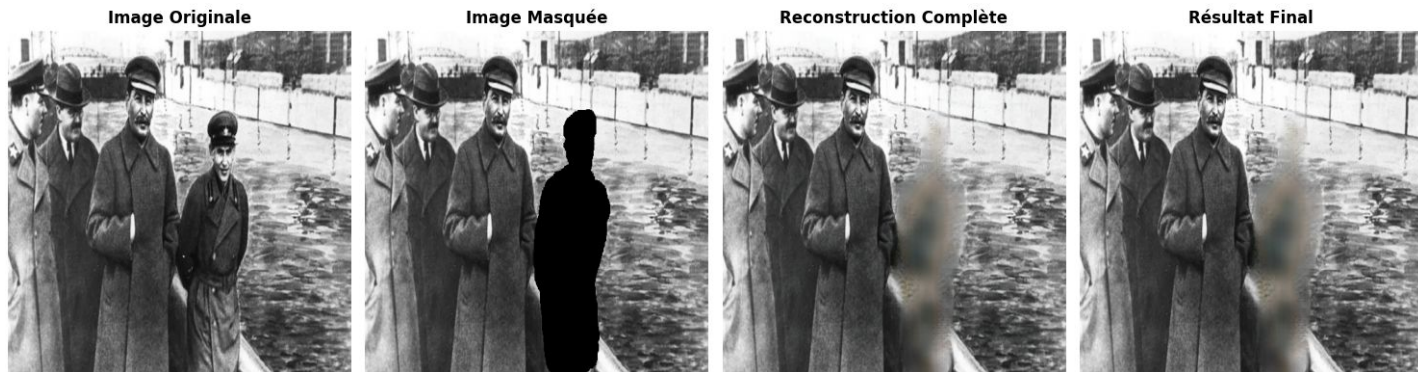
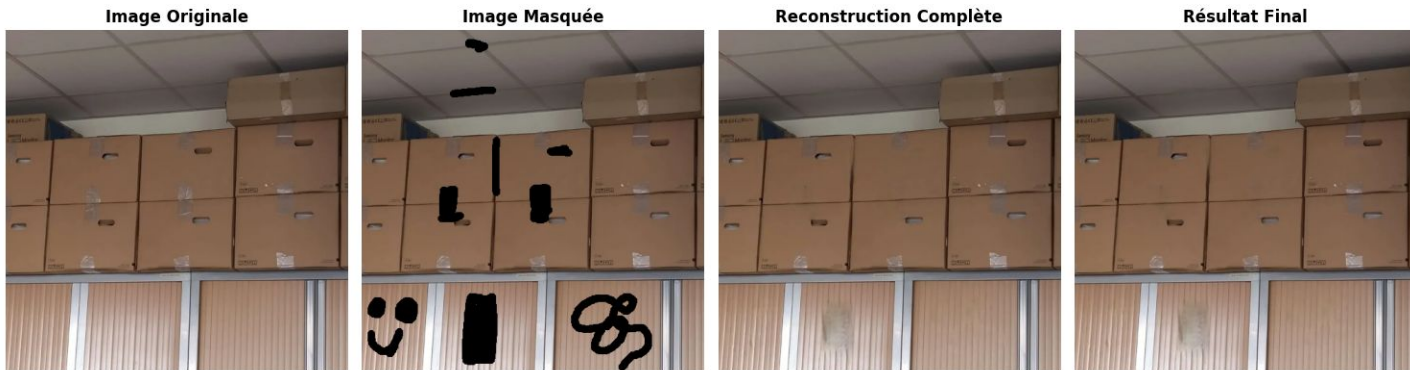
1

Smoothing

72

Courbes approximatives

U-Net GAN



PEN-Net GAN



Image Originale

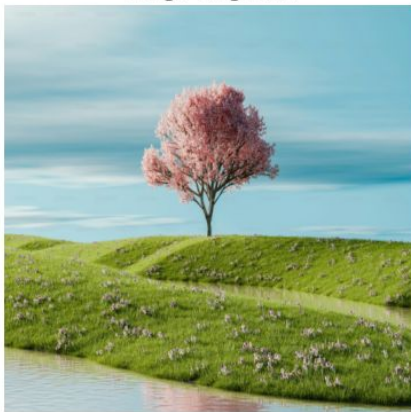
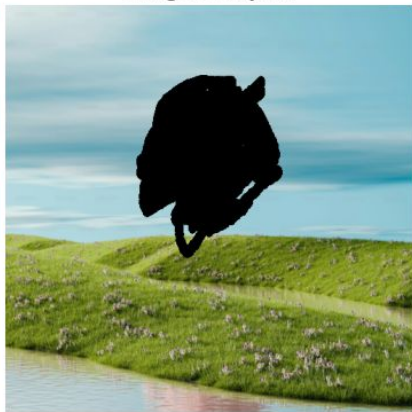


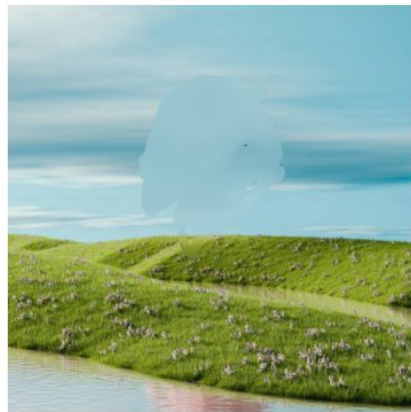
Image Masquée



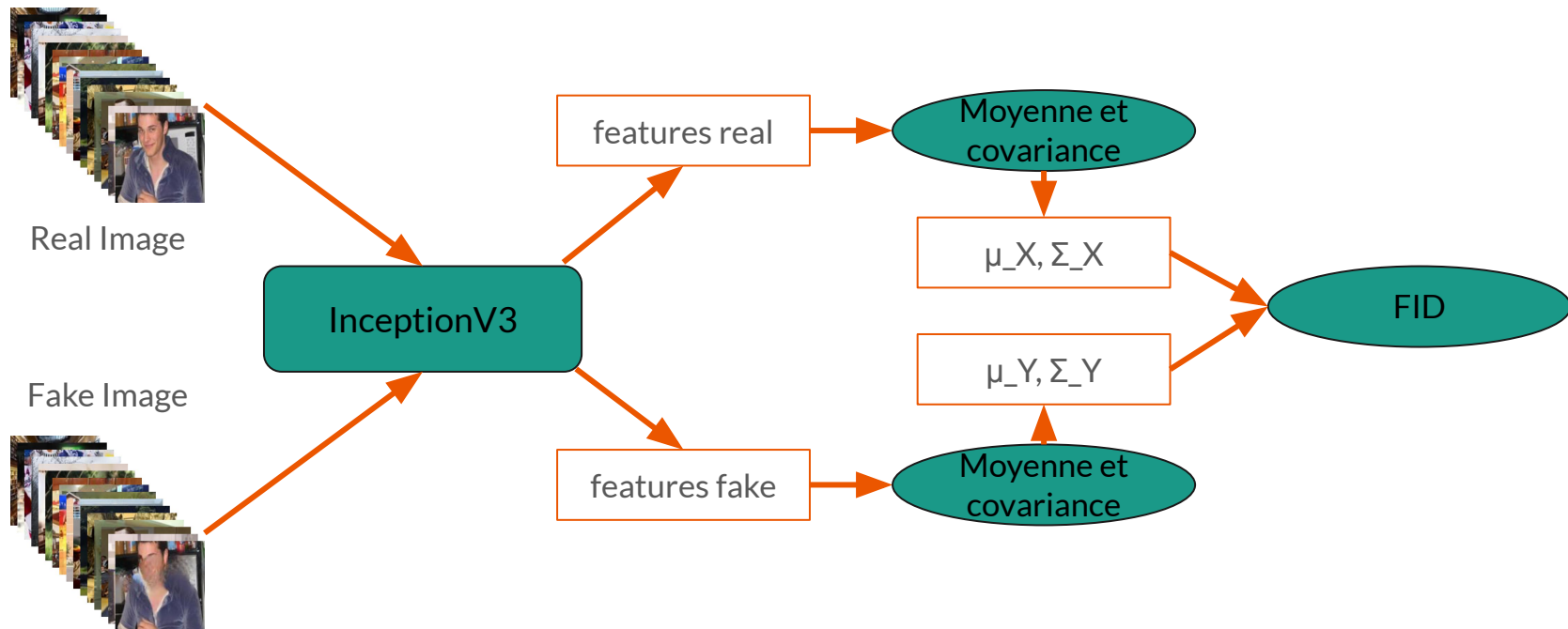
Reconstruction Complète



Résultat Final

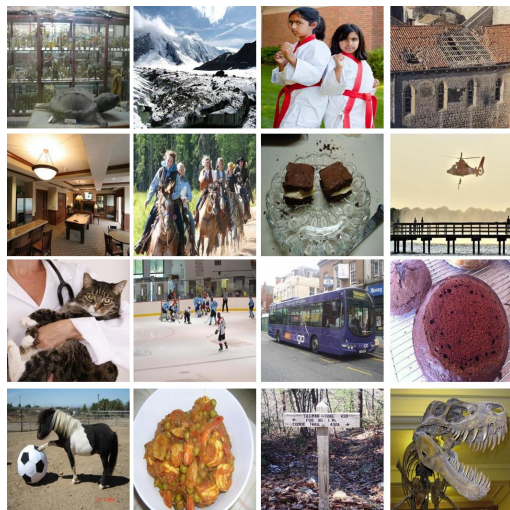


FID Test



$$\text{FID}(X, Y) = \|\mu_X - \mu_Y\|_2^2 + \text{Tr} \left(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{\frac{1}{2}} \right)$$

FID Test



Extrait de 10000 images réduite en
128x128 du dataset
Places-2_MIT_Dataset (validation set)

FID Test



Notre PEN-Net
Entraîner sur Places
8 epoch
FID score: 54.8043

GAN-Unet
Entraîner sur Places
18 epoch
FID score: 9.1791

GAN-Unet
Entraîner sur Cifar10 (32x32)
4 epoch
FID score: 83.6395

2

1

3



Adobe 3000

Adobe 3000

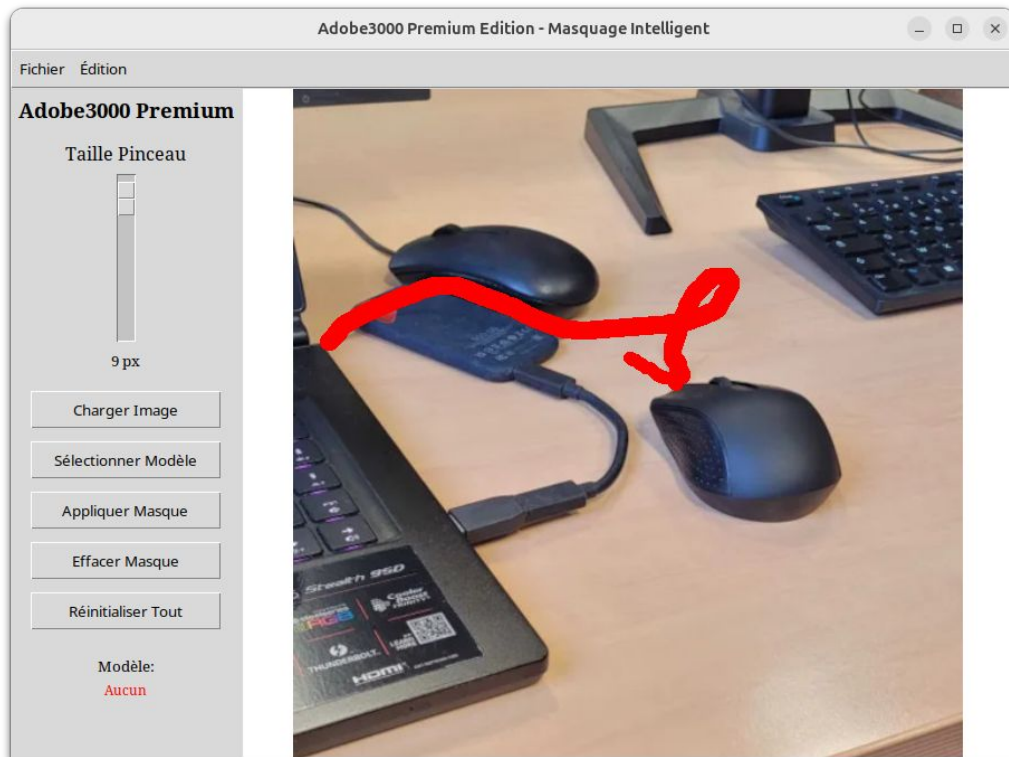
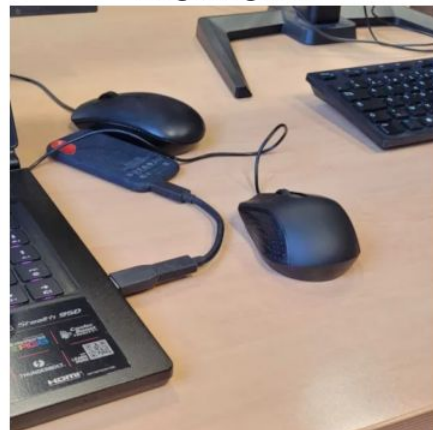
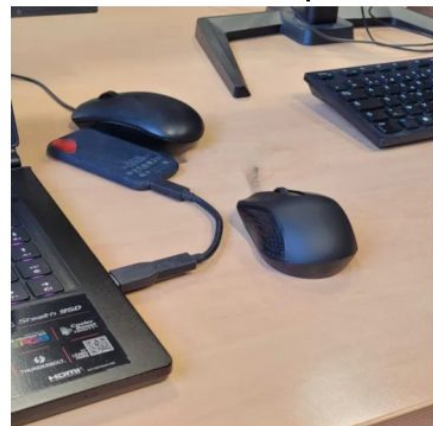


Image Originale



Reconstruction Complète





Perspectives

- Tester avec plus de fonctions de perte
- Pouvoir utiliser Place2 en entier
- Réaliser des test plus poussés sur PEN-Net pour pouvoir combler des zones plus larges et complexes sans flou
- Expérimenter avec les modèles de diffusions pour générer les zones manquantes



Sources

- Learning Pyramid-Context Encoder Network for High-Quality Image Inpainting (2019, cité 661 fois) (PEN-Net)
<https://arxiv.org/pdf/1904.07475>
- Resolution-robust Large Mask Inpainting with Fourier Convolutions (2021, cité 1409 fois)
<https://arxiv.org/abs/2109.07161>
- Github implémentation MAE
<https://github.com/IcarusWizard/MAE>
- Tutoriel implémentation U-Net
<https://medium.com/data-science/cook-your-first-u-net-in-pytorch-b3297a844cf3>
- Github implémentation de PEN-Net
<https://github.com/researchmm/PEN-Net-for-Inpainting>



Sources

- A Review of Image Inpainting Methods Based on Deep Learning (2023, cité 47 fois)
<https://www.mdpi.com/2076-3417/13/20/11189>
- Generative Image Inpainting With Contextual Attention (2018, cité 3262 fois)
<https://arxiv.org/abs/1801.07892v2>
- EdgeConnect: Generative Image Inpainting with Adversarial Edge Learning, (2019, cité 1057 fois)
<https://arxiv.org/abs/1901.00212>
- Towards Context-Stable and Visual-Consistent Image Inpainting
<https://arxiv.org/html/2312.04831v2>
- Masked Autoencoders: A Simple Yet Powerful Approach to Self-Supervised Vision Learning
<https://medium.com/@jimcanary/masked-autoencoders-a-simple-yet-powerful-approach-to-self-supervised-vision-learning-0ec9dc849dd2>



Merci de votre attention !